# Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

## Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

## 1 Inconsistent convergence in Hilbert space

**Files.**

- File: `briefs/InconsistentConvergence/InconsistentConvergence.lean`
- Library: `InconsistentConvergence`
- Imports: `Mathlib.Analysis.Normed.Lp.lpSpace`

**Setting.** $H = \ell^2(\mathbb{N})$ (`lp (fun _ : ℕ => ℂ) 2`), with $|n\rangle = \delta_n$ the oscillator eigenstates and

$$|e^j\rangle = \sqrt{\tfrac{j-1}{j}}\;|0\rangle + \sqrt{\tfrac{1}{j}}\;|j\rangle\,.$$

**Results.**

| Brief | Lean |
|---|---|
| eq. (2): $\|e^j - \|0\rangle\|^2 = \left(\sqrt{(j-1)/j} - 1\right)^2 + 1/j$ | `norm_sub_ground_sq` |
| $e^j \to \|0\rangle$ in $\ell^2$ | `tendsto_e` |
| eq. (3): $\sum_n n^2 \|e^j_n\|^2 = j \to \infty$ | `secondMoment_e`, `tendsto_secondMoment_e` |
| the $\|j\rangle$ alone have no limit in $\ell^2$ | `not_cauchySeq_level`: $\| \|j\rangle - \|k\rangle \|^2 = 2$ for $j \neq k$ |
| eq. (4): $\langle e^j \| N \| e^j \rangle = 1$ for $j \geq 1$ | `numberExp_e` |
| eq. (5), the inconsistency | `not_continuousAt_numberExp`: $\langle N \rangle$ is not continuous at $\|0\rangle$ in $\ell^2$ |

**Differences from the brief.**

- Indexing. Proposition 1 indexes sequences from 1, with the ground term at $i = 1$. The formalization indexes from 0, matching the oscillator of Section 3. The divergence is the same; with the brief's indexing the sum is $(j-1)/j + j$ instead of $j$.
- No limit in $S(\mathbb{N})$. What is proved is that the seminorm $\sum_n n^2 |x_n|^2$ is unbounded along $e^j$, which is the brief's argument. The Schwartz sequence space $S(\mathbb{N})$ and its topology are not constructed.
- $\langle N \rangle$ is defined on all of $\ell^2$ as the weighted sum $\sum_n n\,|x_n|^2$, which has a default value outside the domain of $N$. The discontinuity is still meaningful, because every $e^j$ is finitely supported.

**Problems found in the brief.** No mathematical problems. Minor: the distance is written $d(x, y) = \sum_i (x_i - y_i)^2$, which is the squared distance and, for complex entries, should read $\sum_i |x_i - y_i|^2$.